← Back **Technology: Internets and Packets**
Graded Assignment • 30 min

Due Mar 16, 11:59 PM CET

1. Common Link Layer technologies are… (Choose all that apply) — 1 point

- [ ] VCRs
- [x] DSL
- [x] Optical
- [ ] DVD players

2. When do wireless devices receive their serial numbers (i.e. MAC or Ethernet addresses)? — 1 point

- ( ) These numbers are assigned to individual people, and every device they own has the same number
- ( ) Every time they connect to the internet
- ( ) When they pair with a wireless router
- (•) When they are manufactured

3. What does the time taken for a packet to reach a destination usually reflect? — 1 point

- ( ) The type of data the packet makes up
- ( ) How large the total message or data element is
- (•) The speed of light and the distance the packet has to travel
- ( ) How much the individual user sending the information has paid for their internet connection

4. How do wireless devices operating on a shared network determine when to send information so as not to incur chaos? — 1 point

- ( ) They send requests to all other devices on the network, and wait to receive permission before transmitting data.
- (•) They listen to the sound on the current network, and send information when it is quiet.
- ( ) They chart energy usage, and send information when the numbers are low
- ( ) There is only one link to the network, and only one wireless device can connect at a time, so they are physically prevented from sending information unless it is their turn.

5. What is the concern when deciding which device sends information next on Ethernet? — 1 point

- (•) Ensuring fairness - that one type of device, data, or user is not preferred over others.
- ( ) Sending the most urgent emails before less important messages (like Farmville notifications)
- ( ) Prioritizing the customers who purchase premium internet plans
- ( ) Discouraging the sending of large messages by delaying their transmission in favor of smaller, faster messages

6. What is the maximum possible number of hops a packet can take to try to reach their destination (the so-called "Time To Live" functionality of packets)? — 1 point

- ( ) 4
- ( ) 500
- ( ) 150
- (•) 255

7. What are Router Tables? — 1 point

- ( ) A linked trio of routers that manages incoming, outgoing, and within-network data transmissions.
- (•) Dynamic lists of directions for where and how to direct packets
- ( ) An electrically enhanced table that, when you place a router on it, will increase your network speed
- ( ) Huge banks of routers, housed by Google, that direct Internet traffic

8. What are the layers, and in what order do we structure them? — 1 point

- ( ) Internetwork Layer
  Application Layer
  Link Layer
  Transport Layer
- (•) Application Layer
  Transport Layer
  Internetwork Layer
  Link Layer
- ( ) Transport Layer
  Packet Layer
  Visual Layer
  Link Layer
- ( ) Link Layer
  Map Layer
  Social Media Layer
  Application Layer

9. What is the Internet Protocol Layer responsible for? — 1 point

- ( ) Moving the packet onto the link
- (•) Getting a packet to a specific network address
- ( ) Being 100% reliable
- ( ) Managing the order of data transmission from multiple computers on a wireless network

10. How is an IP address determined? — 1 point

- ( ) By the date in which the owner first got an email account
- (•) Geographically
- ( ) According to product manufacturing date
- ( ) By the hour in which the computer was most recently turned on

11. The prefix of an IP address determines what? — 1 point

- (•) The network that it belongs to
- ( ) The owner of the computer
- ( ) The brand of computer
- ( ) The default web browser installed

12. What is the Link Layer responsible for? — 1 point

- ( ) Moving the packet to the final destination
- (•) Moving the data onto a single link
- ( ) Storing each packet until it has been acknowledged for delivery
- ( ) Reporting which packets successfully arrived at their destination

13. Is it possible to track a packet's journey across the network? — 1 point

- ( ) Yes, using a service called 'packetfind' that tracks the transmission of all packets across the Internet.
- (•) Yes, using a technique called 'traceroute' which tracks the packets that are returned due to transmission failure.
- ( ) Yes, using RIP (Router Information Protocol) which tracks the packets that successfully arrive at their destination.
- ( ) No, packets cannot be tracked.